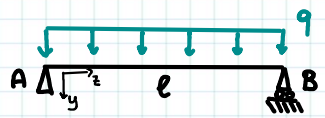
ES 1

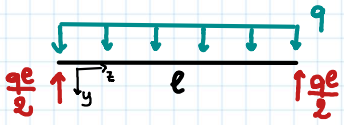


M, T, N continui su tutto ℓ

1° step: devo conoscere le REAZ. VINCOLARI

2° step: calcolo gli andamenti M, T, N.

METODO 1



$$\begin{cases} \dfrac{dN}{dz} = 0 & N = \text{cost} = 0 \\ N(z=0) = 0 \end{cases}$$

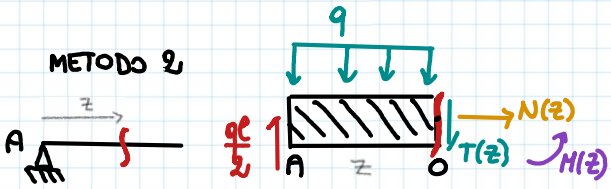
$$\begin{cases} \dfrac{dT}{dz} = -q & T = -qz + C \quad T = \dfrac{q\ell}{2} - qz \\ T(z=0) = \dfrac{q\ell}{2} \end{cases}$$

$\dfrac{q\ell}{2}$ $z=0$

$-\dfrac{q\ell}{2}$ $z=\ell$

$$\begin{cases} \dfrac{dM}{dz} = T(z) = -qz + \dfrac{q\ell}{2} & M = -q\dfrac{z^2}{2} + \dfrac{q\ell}{2}z + C \\ M(z=0) = 0 & C = 0 \quad M(z) = -q\dfrac{z^2}{2} + \dfrac{qz\ell}{2} \end{cases}$$

METODO 2



sez. a distanza z da A

Eq orizz di AO $N(z) = 0$

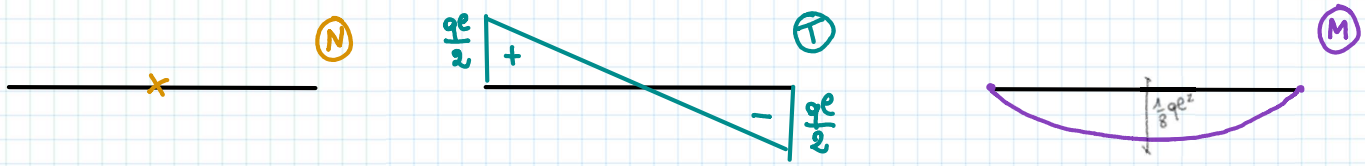
Eq verticale di AO $T(z) - \dfrac{q\ell}{2} + qz = 0$

$T(z) = \dfrac{q\ell}{2} - qz$

Eq. momento in O $M(0) = M(z) + q\dfrac{z^2}{2} - q\dfrac{\ell}{2}z = 0$

$M(z) = q\dfrac{\ell z}{2} - q\dfrac{z^2}{2}$

Rappresentazione grafica

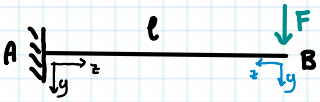


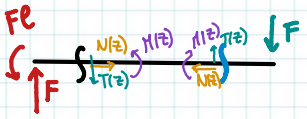
Massimo?

$\dfrac{dM(z)}{dz} = 0 \quad \dfrac{q\ell}{2} - qz = 0 \quad z = \dfrac{\ell}{2}$

$M\left(\dfrac{\ell}{2}\right) = \dfrac{q\ell^2}{4} - \dfrac{q\ell^2}{8} = \dfrac{q\ell^2}{8}$

ES 2





Metodo 1:

$$\begin{cases} \dfrac{dN}{dz} = 0 & N = \text{cost} = 0 \\ N(z=0) = 0 \end{cases}$$

$$\begin{cases} \dfrac{dT}{dz} = 0 & T = \text{cost} = F \\ T(z=0) = F \end{cases}$$

$$\begin{cases} \dfrac{dM}{dz} = T = F & M = Fz + C \quad M = -F\ell + Fz \\ M(z=0) = -F\ell & C = -F\ell \end{cases}$$

Metodo 2 (sist zy)

$N(z) = 0$

$T(z) - F = 0 \quad T(z) = F$

$M(0) = M(z) + F\ell - Fz = 0$

$M(z) = Fz - F\ell \longrightarrow z=0 \quad M = -F\ell$

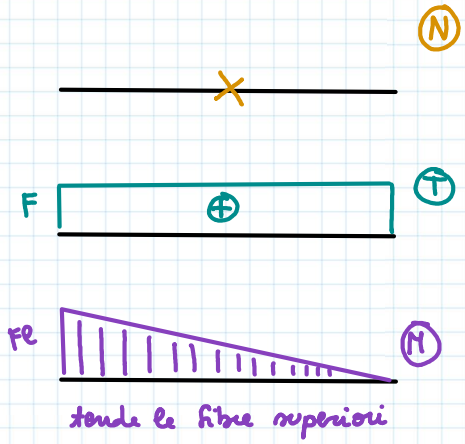
$\longrightarrow z = \ell \quad M = 0$

sist. zy

$N(z) = 0$

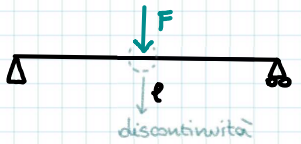
$T(z) - F = 0 \quad T(z) = F$

$M(0) = M(z) + Fz = 0 \quad M(z) = -Fz$ → $M(z=0) = 0$, → $M(z=\ell) = -F\ell$

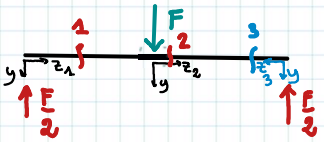


tende le fibre superiori

ES 3



NB Una F concentrata causa un SALTO nel diagramma di TAGLIO pari all'entità del modulo della forza.



① $N(z_1)=0$

$T(z_1)=\frac{F}{2}$

$M(z_1)-\frac{F}{2}z_1=0 \quad M(z_1)=\frac{F}{2}z_1$

② $N(z_2)=0$

$T(z_2)+F-\frac{F}{2}=0 \quad T(z_2)=-\frac{F}{2}$

$M(z_2)+Fz_2-\frac{F}{2}\left(\frac{\ell}{2}+z_2\right)=0$

$M(z)=\frac{F\ell}{4}-\frac{F}{2}z_2$

③ $N(z_3)=0$

$T(z_3)+\frac{F}{2}=0 \quad T(z_3)=-\frac{F}{2}$

$M(z_3)=\frac{F}{2}z_3$

Integrazione

① $N(z)=0$

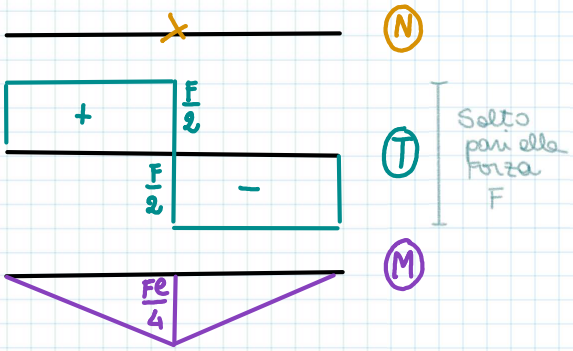
$\begin{cases}\frac{dT}{dz_1}=0 \quad T=\frac{F}{2}\\ T(z_1)=\frac{F}{2}\end{cases}$

$\begin{cases}\frac{dM}{dz_1}=T=\frac{F}{2} \quad M(z_1)=\frac{F}{2}z_1+C\\ M(z_1=0)=0 \quad M(z_1)=\frac{F}{2}z_1\end{cases}$

② $N(z_2)=0$

$\begin{cases}\frac{dT}{dz_2}=0 \quad T(z_2)=-\frac{F}{2}\\ T(z_2=0)=-F+\frac{F}{2}\end{cases}$

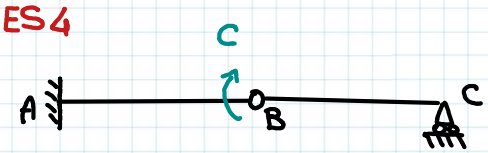
$\begin{cases}\frac{dM}{dz_2}=-\frac{F}{2} \quad M=-\frac{F}{2}z_2+\frac{F\ell}{4}\\ M(z_2=0)=\frac{F\ell}{4}\end{cases}$

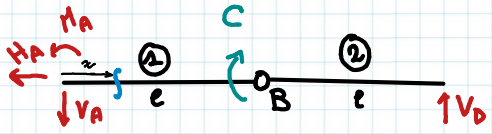


ATTENZIONE

Se integro in z_3, cambia $\frac{dM}{dz}=-T$ (da eq. del concio)

ES 4



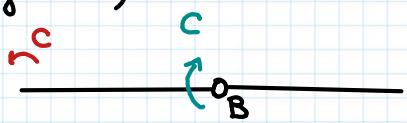


Eq^{ne} ausiliaria in B del corpo 2

$M(B)=V_D\cdot\ell=0 \quad V_D=0 \Rightarrow V_A=0$ (eq globale ↑)

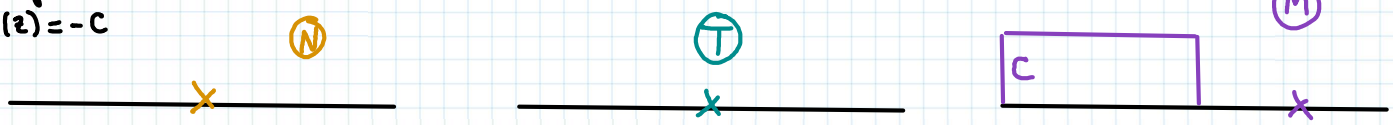
$H_A=0$ (eq globale →)

$M(A)=M_A-C=0 \quad M_A=C$

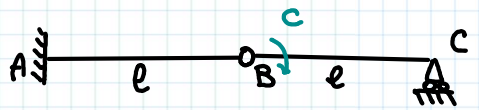


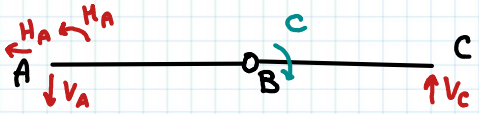
Diagrammi:

$M(z)=-C$



ES 5



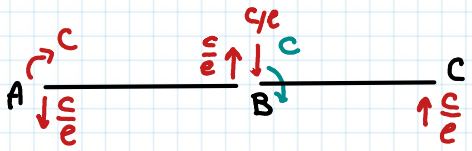


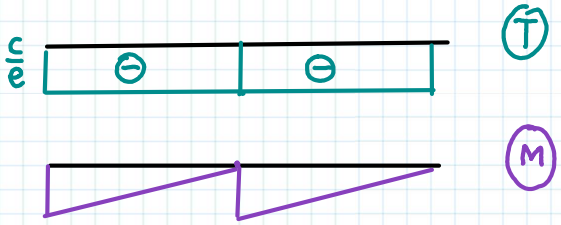
Eq.^{ne} ausiliaria corpo 2

$\int M(B) = V_C \cdot \ell - C = 0 \quad V_C = \frac{C}{\ell} = V_A$ eq ↑ globale

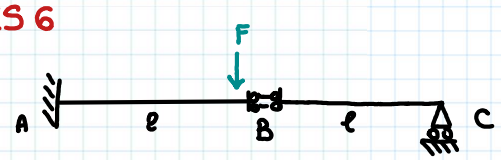
$H_A = 0$ eq. orizz globale

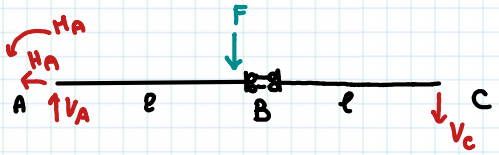
$\int M(A) = M_A - C + V_C \cdot 2\ell = 0 \quad M_A - C + 2C = 0 \quad M_A = -C$





ES 6





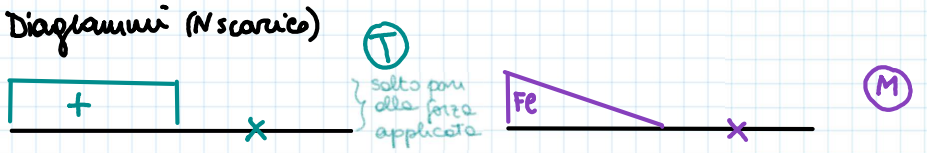
Eq.^{ne} ausiliaria corpo 2 $\quad V_C = 0$

Eq. globale ↑ $\quad V_A = F$

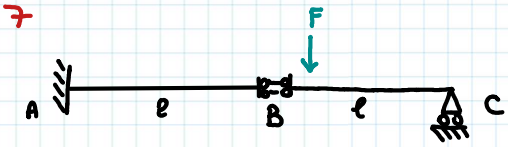
$H_A = 0$

$M_A = F\ell$

Diagrammi (N scarico)



ES 7



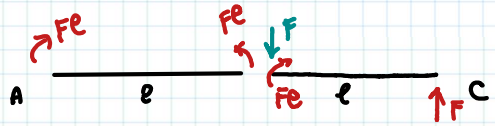
Eq^{ne} ausiliaria corpo 2 : $V_C = -F$

Eq. globale ↑ $V_A = 0$

→ $H_A = 0$

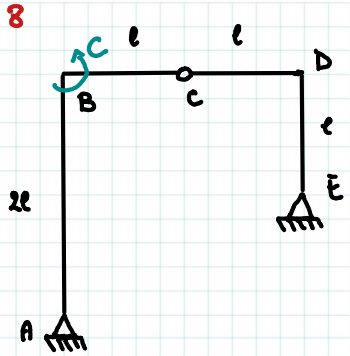
$\int M(A) = M_A - F\ell - V_C 2\ell = 0$

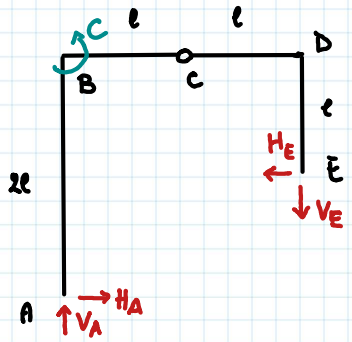
$M_A - F\ell + 2F\ell = 0 \quad M_A = -F\ell$



Diagrammi (N scarico)

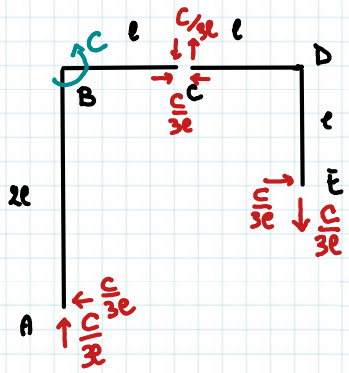
ES 8



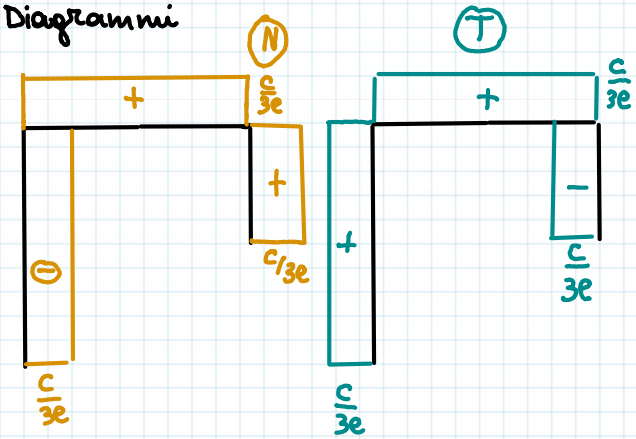


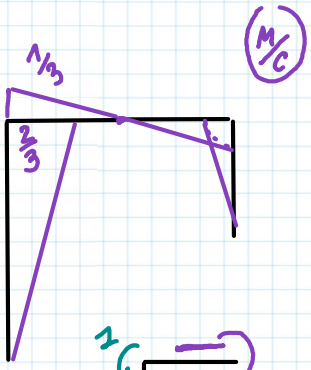
$$\begin{cases} H_A - H_E = 0 & H_A = -\frac{C}{3l} \\ V_A - V_E = 0 & V_A = \frac{C}{3l} \\ \int M(A) = C + H_E l - V_E 2l = 0 & C - V_E l - 2V_E l = 0 \\ \int M(C) = -H_E l - V_E l = 0 & H_E = -V_E \end{cases}$$

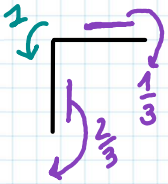
$$V_E = \frac{C}{3l} \qquad H_E = -\frac{C}{3l}$$



Diagrammi

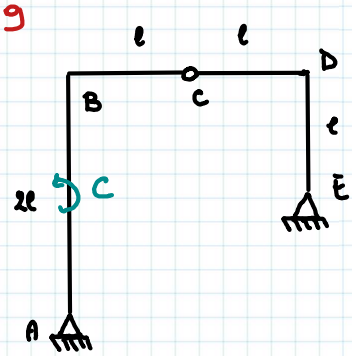


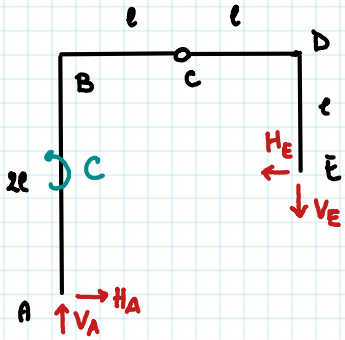




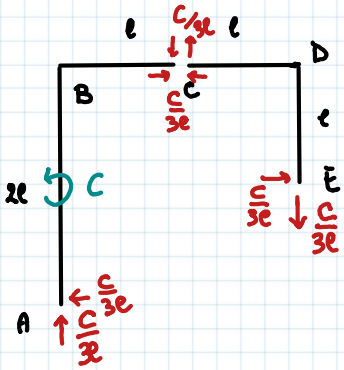
NODO EQUILIBRATO

ES 9

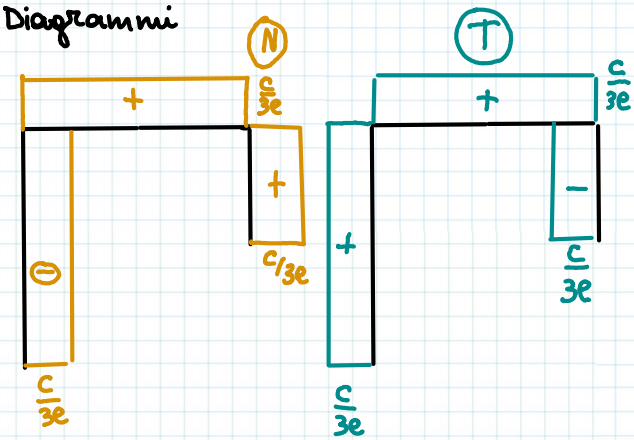


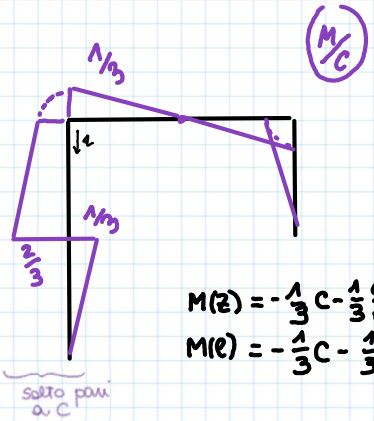


Stesse reazioni di prima (es 8)



Diagrammi

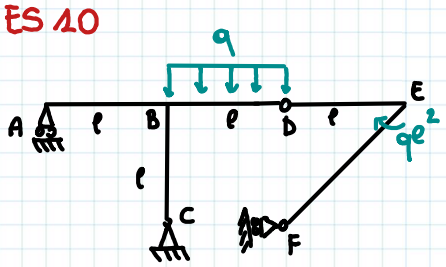


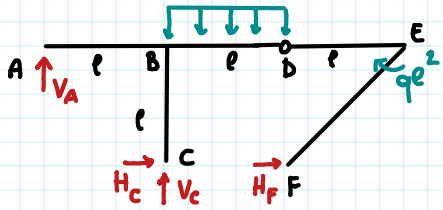


$$M(z) = -\frac{1}{3}C - \frac{1}{3}\frac{C}{l}\cdot z$$

$$M(l) = -\frac{1}{3}C - \frac{1}{3}C = -\frac{2}{3}C$$

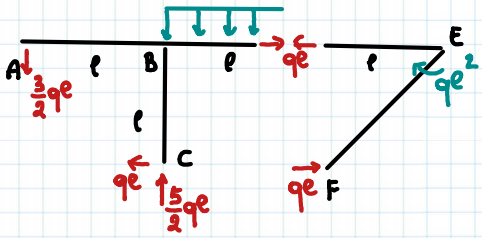
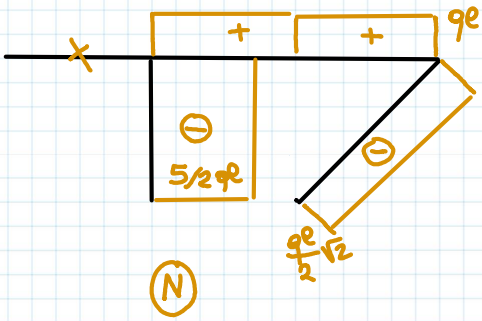
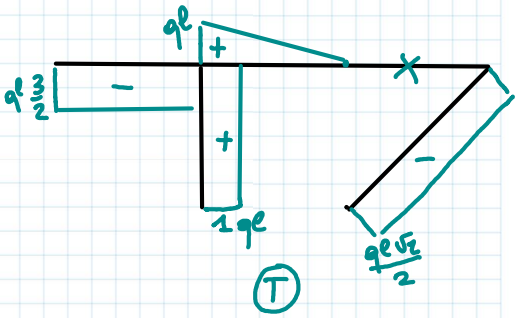
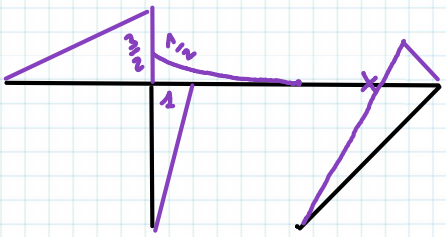
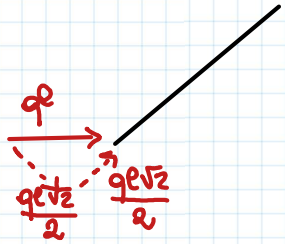
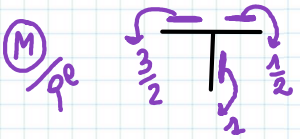
ES 10





$H_C + H_F = 0 \qquad H_C = -q\ell$

$V_A + V_C - q\ell = 0 \qquad V_C = -V_A + q\ell = \frac{5}{2}q\ell$

$\circlearrowleft M(C) = -V_A\,\ell - \frac{q\ell^2}{2} - q\ell^2 = 0 \qquad V_A = -\frac{3}{2}q\ell$

② $\circlearrowleft M(D) = H_F \cdot \ell - q\ell^2 = 0 \qquad H_F = q\ell$